Lecture 23 (*April 14*)

Reading: Section 6.2
Viewing: Lecture 21

In the previous class, we saw how to compute the eigenvalues and eigenvectors of an $n \times n$ matrix A. Specifically, the eigenvalues are the roots of the characteristic polynomial:

$$p(\lambda) = \det(A - \lambda I) = 0$$

Since the characteristic polynomial has degree n, it has at most n roots, therefore there are at most n eigenvalues. But there could be fewer than n eigenvalues, for example the matrix:

$$A = \begin{bmatrix} 5 & -3 \\ 0 & 5 \end{bmatrix}$$

has characteristic polynomial

$$p(\lambda) = (\lambda - 5)^2$$

which has $\lambda = 5$ as a double root. In this case, we say that $\lambda = 5$ has **algebraic multiplicity** 2, or that the matrix A has repeated (or equal) eigenvalues. In other words, we still think of A has having two eigenvalues, but they are both equal to 5. With this in mind, we have:

Fact 18. *An $n \times n$ matrix has exactly n eigenvalues, although some could be repeated, and some could be complex numbers (more on complex numbers later).*

So what can we say about eigenvectors? First of all, eigenvectors corresponding to different eigenvalues are always linearly independent. For example, if:

$$A\boldsymbol{v}_1 = \lambda_1 \boldsymbol{v}_1 \qquad \text{and} \qquad A\boldsymbol{v}_2 = \lambda_2 \boldsymbol{v}_2$$

for $\lambda_1 \neq \lambda_2$, then there can be no linear relation between $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ (argument: if there were such a relation, then we would have $\boldsymbol{v}_1 = c\boldsymbol{v}_2$ for some scalar c, and applying A would give us $\lambda_1 \boldsymbol{v}_1 = A\boldsymbol{v}_1 = cA\boldsymbol{v}_2 = c\lambda_2 \boldsymbol{v}_2 = \lambda_2 \boldsymbol{v}_1$, which would force $\lambda_1 = \lambda_2$). So the best case is when the n eigenvalues of the matrix are all distinct numbers, because then there would exist n linearly independent eigenvectors, one for each eigenvalue. These eigenvectors would form a basis of $\mathbb{R}^n$.

Fact 19. *An $n \times n$ matrix A with n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ is diagonalizable. Specifically:*

$$A = V \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix} V^{-1} \tag{191}$$

where V is the matrix whose columns are $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$, the eigenvectors corresponding to $\lambda_1, \ldots, \lambda_n$.

A couple of remarks: if the eigenvalues are all distinct, then the eigenvectors are really only defined up to constant multiple. So the matrix V in (191) is not unique, but it depends on a choice of constant multiples of the eigenvectors. Secondly, the fact that the eigenvectors are linearly independent

(which was discussed right before Fact 19) is crucial, because otherwise the matrix V would have rank $< n$ and would not be invertible and the right-hand side of formula (191) would not make sense.

In practice, how to diagonalize (i.e. to write as in (191)) a matrix A? As explained in Fact 19, the way to do so is to find the eigenvalues and eigenvectors of A. Say we're looking at the matrix:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \tag{192}$$

To find its eigenvalues, construct the characteristic polynomial:

$$p(\lambda) = \lambda^2 - \lambda \cdot \operatorname{tr} A + \det A = \lambda^2 - \lambda - 1$$

(here we're applying the shortcut formula (190), which only holds in the 2×2 case). Its roots are:

$$\lambda_1 = \frac{1+\sqrt{5}}{2} \qquad \text{and} \qquad \lambda_2 = \frac{1-\sqrt{5}}{2}$$

Let us now compute eigenvectors corresponding to these two eigenvalues:

$$\begin{aligned} A\boldsymbol{v}_1 = \frac{1+\sqrt{5}}{2}\boldsymbol{v}_1 \quad &\Rightarrow \quad \left(A - \frac{1+\sqrt{5}}{2}I\right)\boldsymbol{v}_1 = 0 \quad \Rightarrow \quad \boldsymbol{v}_1 \in N\left(A - \frac{1+\sqrt{5}}{2}I\right) \\ A\boldsymbol{v}_2 = \frac{1-\sqrt{5}}{2}\boldsymbol{v}_2 \quad &\Rightarrow \quad \left(A - \frac{1-\sqrt{5}}{2}I\right)\boldsymbol{v}_2 = 0 \quad \Rightarrow \quad \boldsymbol{v}_2 \in N\left(A - \frac{1-\sqrt{5}}{2}I\right) \end{aligned}$$

To work out the nullspace of $A - \frac{1+\sqrt{5}}{2}I$, put it in row echelon form:

$$A - \frac{1+\sqrt{5}}{2}I = \begin{bmatrix} \frac{1-\sqrt{5}}{2} & 1 \\ 1 & \frac{-1-\sqrt{5}}{2} \end{bmatrix} \rightsquigarrow \begin{bmatrix} \frac{1-\sqrt{5}}{2} & 1 \\ 0 & 0 \end{bmatrix}$$

So the eigenvector $\boldsymbol{v}_1 = \begin{bmatrix} x \\ y \end{bmatrix}$ needs to satisfy:

$$\begin{bmatrix} \frac{1-\sqrt{5}}{2} & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \quad \Leftrightarrow \quad y + \frac{1-\sqrt{5}}{2}x = 0$$

and therefore we may take $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ \frac{-1+\sqrt{5}}{2} \end{bmatrix}$. Similarly, $\boldsymbol{v}_2 = \begin{bmatrix} 1 \\ \frac{-1-\sqrt{5}}{2} \end{bmatrix}$, so we have:

$$A = V \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix} V^{-1} \qquad \text{where} \qquad V = \begin{bmatrix} 1 & 1 \\ \frac{-1+\sqrt{5}}{2} & \frac{-1-\sqrt{5}}{2} \end{bmatrix} \tag{193}$$

So we've done all this work, but you might ask: what are applications of diagonalization? Well, here's a classic one. Let's say you want to obtain an explicit formula for the Lucas numbers (close cousins of the Fibonacci numbers considered in the textbook), which are defined by the conditions:

$$L_0 = 2, \qquad L_1 = 1, \qquad \text{and the recursive relation} \qquad L_{n+1} = L_n + L_{n-1} \tag{194}$$

for all $n \geq 1$. One way would be to consider the vectors:

$$\boldsymbol{a}_n = \begin{bmatrix} L_{n+1} \\ L_n \end{bmatrix}$$

and to observe that the conditions (194) can be interpreted as the matrix identities:

$$\boldsymbol{a}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{a}_n = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \boldsymbol{a}_{n-1}$$

for all $n \geq 1$. You can iterate the latter formula and obtain:

$$\boldsymbol{a}_n = A\boldsymbol{a}_{n-1} = A^2\boldsymbol{a}_{n-2} = \cdots = A^n\boldsymbol{a}_0 = A^n \begin{bmatrix} 1 \\ 2 \end{bmatrix} \tag{195}$$

where A is the matrix (192). So computing a formula for the Lucas numbers boils down to having an effective method for explicitly computing the matrix powers A^n for all natural numbers n. This is quite straightforward if you kow how to diagonalize A. Specifically, because A can be written in the form (193), we have:

$$A^2 = V \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix} \underbrace{V^{-1}V}_{I} \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix} V^{-1} = V \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^2 & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^2 \end{bmatrix} V^{-1}$$

$$A^3 = V \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix} \underbrace{V^{-1}V}_{I} \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix} \underbrace{V^{-1}V}_{I} \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix} V^{-1} = V \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^3 & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^3 \end{bmatrix} V^{-1}$$

$$\cdots$$

$$A^n = V \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^n & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^n \end{bmatrix} V^{-1}$$

for all integers n. We could plug these formulas in (195) and obtain:

$$\begin{bmatrix} L_{n+1} \\ L_n \end{bmatrix} = \boldsymbol{a}_n = V \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^n & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^n \end{bmatrix} V^{-1} \begin{bmatrix} 1 \\ 2 \end{bmatrix} =$$

$$= \begin{bmatrix} 1 & 1 \\ \frac{-1+\sqrt{5}}{2} & \frac{-1-\sqrt{5}}{2} \end{bmatrix} \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^n & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^n \end{bmatrix} \frac{1}{-\sqrt{5}} \begin{bmatrix} \frac{-1-\sqrt{5}}{2} & -1 \\ \frac{1-\sqrt{5}}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

I'll let you have all the fun in multiplying the matrices on the bottom row above. When you do so, you will finally obtain the formula for the Lucas numbers:

$$L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n \tag{196}$$

The example we have just seen generalizes to the following principle:

$$\text{if} \quad A = V \begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix} V^{-1} \quad \text{then} \quad A^k = V \begin{bmatrix} \lambda_1^k & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n^k \end{bmatrix} V^{-1} \tag{197}$$

This is computationally very important, because if you want to compute the k–th power of an $n \times n$ matrix efficiently for some really large number k, it is faster to just diagonalize A and apply formula (197). Also, this formula allows us to describe the behavior of A^k as $k \to \infty$: its growth rate is dominated by the size of the k–th power of the largest eigenvalue of A.

Remark. *The Lucas numbers L_n grow like $\left(\frac{1+\sqrt{5}}{2}\right)^n$ as $n \to \infty$. This makes sense, since the other eigenvalue $\frac{1-\sqrt{5}}{2}$ lies between -1 and 1, and so its n–th powers will converge to 0 as $n \to \infty$.*

Geometrically, suppose you want to compute the value of A^k applied to any vector $\boldsymbol{v}$? Just decompose this vector as a linear combination of the eigenvectors:

$$\boldsymbol{v} = c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n \quad \Rightarrow \quad A^k\boldsymbol{v} = c_1A^k\boldsymbol{v}_1 + \cdots + c_nA^k\boldsymbol{v}_n$$

Since $A\boldsymbol{v}_i = \lambda_i\boldsymbol{v}_i \Rightarrow A^k\boldsymbol{v}_i = \lambda_i^k\boldsymbol{v}_i$ for all i (on account of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ being eigenvectors), we get:

$$\boxed{A^k\boldsymbol{v} = c_1\lambda_1^k\boldsymbol{v}_1 + \cdots + c_n\lambda_n^k\boldsymbol{v}_n} \tag{198}$$

We conclude that the powers of A scale by powers of the eigenvalues $\lambda_1, \ldots, \lambda_n$ in the direction of the eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$, and this gives an effective way to compute A^k times any vector.

Finally, let us mention an easy, but fundamental thing: **the eigenvalues of a diagonal matrix are the entries on the diagonal**. This is because the roots of the polynomial:

$$\det\left(\begin{bmatrix} d_1 & 0 & \ldots & 0 \\ 0 & d_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & d_n \end{bmatrix} - \lambda \cdot I\right) = \det\begin{bmatrix} d_1-\lambda & 0 & \ldots & 0 \\ 0 & d_2-\lambda & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & d_n-\lambda \end{bmatrix} = (d_1-\lambda)\ldots(d_n-\lambda)$$

are precisely $d_1, \ldots, d_n$. Applying this to (191), we see that the matrices A and $\mathrm{diag}_{\lambda_1,\ldots,\lambda_n}$ have the same eigenvalues. This is true for any pair of similar matrices:

$$\textbf{if } A = VBV^{-1} \textbf{ then } A \textbf{ and } B \textbf{ have the same eigenvalues} \tag{199}$$

for any invertible matrix V. However, A and B will in general have different eigenvectors, and these two bases of eigenvectors will be connected by the change of basis matrix V in formula (199):

$$V\boldsymbol{w} \text{ is an eigenvector of } A \quad \Leftrightarrow \quad \boldsymbol{w} \text{ is an eigenvector of } B$$